Name of the Student: _____ **Roll no:** _____

CLASS: - T.E (CSE) **Division:** **Course: - PCC305CSE- Machine Learning Lab**

Experiment No. 02

**** Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) ****

Marks: /10

Date of Performance: / /2026 **Sign with Date:**

Aim: To observe how feature reduction impacts classification accuracy and training speed.

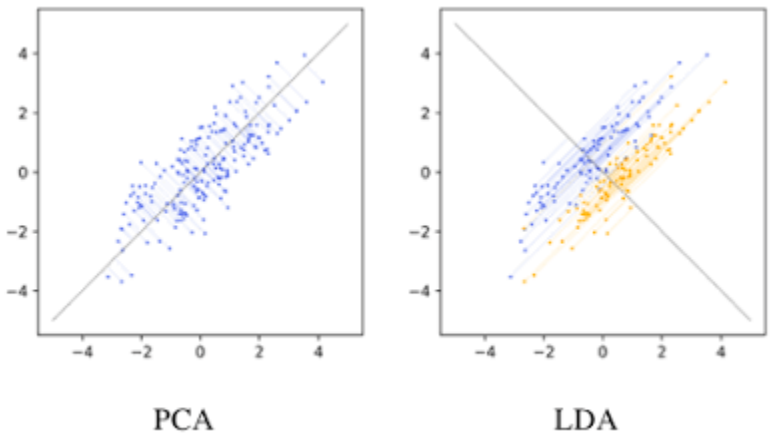
Lab Objective: To apply basic feature engineering techniques to observe how feature reduction impacts classification accuracy and training speed.

Problem Statement:

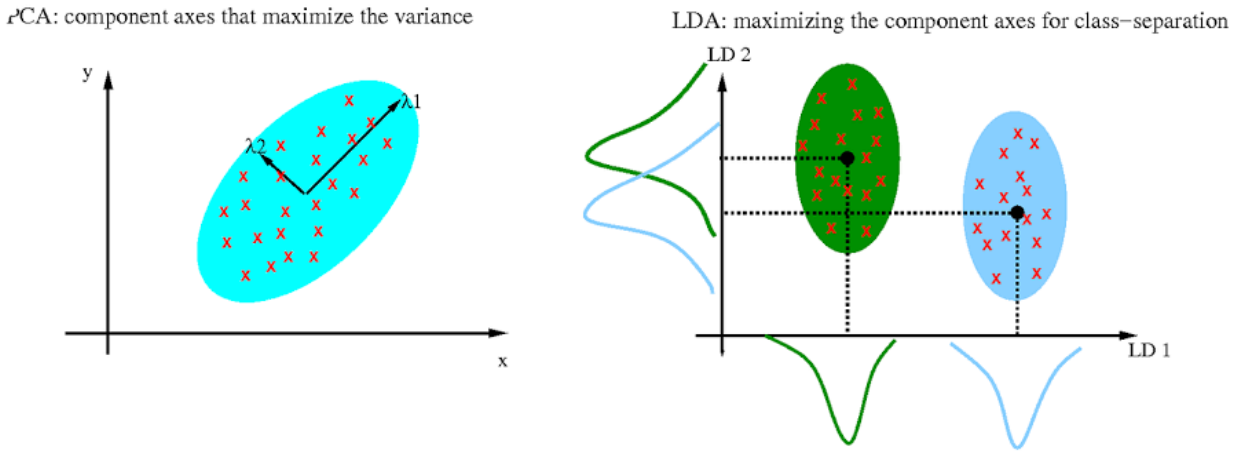
To apply Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) on an open-access high-dimensional dataset, and to observe how feature reduction impacts classification accuracy and training speed.

Theory:

We often treat PCA and LDA as identical "feature compression" tools. The key distinction to emphasize is their underlying objective:



PCA (Unsupervised) vs. LDA (Supervised) Projection.



Principal Component Analysis (PCA) — *Unsupervised*

- **Goal:** Maximize Variance.
- **Intuition:** PCA doesn't care about class labels (e.g., Target Y). It looks at feature space (Features X) and finds the directions (principal components) along which the data varies the most. It projects high-dimensional data onto lower-dimensional axes while losing as little total variance as possible.

Linear Discriminant Analysis (LDA) — *Supervised*

- **Goal:** Maximize Class Separability.
- **Intuition:** LDA actively uses class labels (Target Y). It tries to compress features into axes that **maximize distance between different classes** while **minimizing distance within the same class**.
- **Constraint:** LDA can reduce features down to at most K - 1 dimensions, where K is the number of unique target classes.

Dataset Selection

We use the **Wine Dataset** built into scikit-learn

- **Dimensionality:** 13 numerical features (high enough to demonstrate dimensionality reduction without overwhelming students).
- **Classes:** 3 target wine categories (K=3).
- **Open Access:** Built directly into sklearn.datasets—zero setup or external downloads needed.

Questions:

1. Explain the core difference between Principal Component Analysis (PCA) and Linear Discriminant

Analysis (LDA)

2. What is dimensionality reduction?
3. Why can LDA produce only 1 component here?
4. Explain the theoretical reasons why reducing features using PCA or LDA typically leads to faster model training speeds and can, in some cases, improve classification accuracy rather than degrading it.
5. Can dimensionality reduction reduce accuracy?
6. Implement a program to apply Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) on an open-access high-dimensional dataset, and to observe how feature reduction impacts classification accuracy and training speed.

Attendance and Timely Submission (10)	Execution (10)	Observations and Records (10)	Journal Documentation (10)	Oral (10)

Signature of Subject Teacher